

Probabilistic Machine Learning



Challenges (i) Scalability (ii) Reliability



Inference

- **Exact Inference**
 - Closed-form updating: GP, Kalman filter
- **Approximate Inference**
 - MCMC
 - Variational inference
 - Laplace approximation
- **Sequential Inference**
 - Ensemble methods: EnKF, LETKF
 - Particle methods: PF, SMC
 - Variational methods: 4DVar



Models

- **Nonparametric**
 - Gaussian process, Dirichlet process
- **Bayesian Deep Learning**
 - Bayesian neural network, Deep GP
- **Dynamical Systems**
 - State-space model, Hidden Markov model
- **Graphical Models**
 - Bayesian network, Markov random field
- **Generative Models**
 - VAE, Normalizing flow, Diffusion model



Training

- **Objective**
 - (Marginal) log-likelihood
 - ELBO
 - Scoring rule
- **Optimization**
 - Gradient descent
 - EM, quasi-Newton
- **Selection**
 - Model: architecture, prior, kernel
 - Hyperparameter: learning rate, batch size

Applications



Applications



Domains

- Climate • Weather □ Environment • Earth Science □ Finance • Economics
- Healthcare • Biomedical □ Systems Engineering • Control
- Transportation • Mobility □ Cybersecurity • Reliability